BC Calculus I/II
Spring 2024

Instructor: Dr. Anderson Trimm
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Office hours: I am available daily during S&E.

Availability outside of class: All students are welcome to meet me during the office hours listed above without appointment. If you need to meet with me outside of this time, please send me an email to make an appointment in order to ensure my availability.

Class Meeting Information: Mod 2 in A155 on ABCD days.

Attendance: Please refer to the attendance policies in the IMSA 2023–2024 Student Handbook. My expectation is that you come to class prepared every day and are an active participant in all group activities and class discussions.

Text/Materials: Text: Strang, G. and Herman, E. *Calculus, Vols 1 and 2*, 2016.
Graphing Calculator (TI-89 or TI-84 recommended).
Three-ring binder for class notes, worksheets, quizzes, etc.
Laptop computer
You should bring your binder, calculator, and laptop to class daily.

Canvas: I will post copies of all course handouts, assignments and due dates, notifications of upcoming assessments, and so forth on Canvas. Please make sure to **turn notifications on**.

Course Description

BC I/II is the first semester of the two-semester BC Calculus sequence. Topics of study include limits and continuity, differentiation, applications of differentiation, integration, and techniques of integration.

Course Schedule

Here are the major divisions of this course:

- **Preliminaries**
- **Limits and Continuity**
- **Derivatives**
- **Applications of Derivatives**
- **Integration and Integration Techniques**

Course Content

There are 25 learning standards in the course, which fall under the following course level objectives:

Course Level Objectives

1. **Limits (L)**: Explain, calculate, and apply the concept of limits.
2. **Derivatives (D)**: Explain and interpret the meaning of the derivative of a function.
3. **Derivative Calculations (DC)**: Prove and apply theorems to calculate derivatives efficiently.
4. **Derivative Applications (DA)**: Use derivatives to solve real-world application problems.
5. **Integration Techniques (I)**: Explain, calculate, and apply the definite integral.

Learning Standards

Limits (L): Explain, calculate, and apply the concept of limits.

- L.1: I can find the limit of a function at a point using numerical, graphical, and analytical methods.
- L.2: I can find the limits at infinity, infinite limits, and one-sided limits using numerical, graphical, and analytical methods and interpret the corresponding behavior of the function.
- L.3: I can determine where a function is continuous using the definition, and I can classify discontinuities.
- L.4: I can find limits using the squeeze theorem.
- L.5: I can identify limits in indeterminate form and apply L'Hopital's Rule to evaluate them.

Derivatives (D): Explain and interpret the meaning of the derivative of a function.

- D.1: I can find the derivative of a function, both at a point and as a function, using the definition of the derivative.
- D.2: I can find the equation of the tangent line to a function at a point and use the tangent line to estimate nearby values of the function.
- D.3: I can determine where a function is continuous or differentiable given a graph or formula of the function and explain my reasoning.
- D.4: I can use derivative notation correctly, state the units of a derivative, estimate the value of a derivative using difference quotients, and correctly interpret the meaning of a derivative in context.

Derivative Calculations (DC): Prove and apply theorems to calculate derivatives efficiently.

- DC.1: I can compute derivatives of polynomial, exponential, sine, and cosine functions, basic combinations of these (constant multiples, sums, differences).
- DC.2: I can compute derivatives correctly for products, quotients, and composites of functions.
- DC.3: I can calculate higher-order derivatives of a function.
- DC.4: I can compute derivatives correctly using multiple rules in combination.
- DC.5: I can compute the derivatives correctly for logarithmic, trigonometric, and inverse trigonometric functions.
- DC:6: I can compute the derivative of an implicitly-defined function and find the slope of the tangent line to an implicit curve.

Derivative Applications (DA): Use derivatives to solve real-world application problems.

- DA.1: I can use the First and Second Derivative Tests to classify critical points.
- DA.2: I can use the Extreme Value Theorem to find global extrema.
- DA.3: I can use derivatives to graph a function.
- DA.4: I can use derivatives to solve optimization problems.
- DA.5: I can use the Mean Value Theorem to justify conclusions about a function.
- DA.6: I can solve related rates problems.

Integration (I): Integration and techniques of integration.

- I.1: I can compute integrals of polynomial, exponential, sine, and cosine functions, basic combinations of these (constant multiples, sums, differences).
- I.2: I can compute integrals using substitution.
- I.3: I can compute integrals using integration by parts.
- I.4: I can apply the Fundamental Theorem of Calculus.

Grading

Types of Graded Assignments

1. **Problem Sets**: Problem sets will be assigned every 1-2 weeks, and may include problems from the textbook, problems assigned on an online platform such as WebWork or Quest, as well as additional problems assigned on paper. These may go beyond the learning standards and/or include entry-level contest problems.
2. **Checkpoints**: We will have frequent checkpoints throughout the course to assess the 25 learning standards listed above.
3. **Theory Portfolio**: Students will build a portfolio over the course of the semester containing proofs of theorems encountered throughout the course.
4. **Final Exam**: There will be a cumulative final exam covering the 25 learning standards listed above.

The individual types of assignments are graded as follows:

1. **Problem Sets**: These will be graded for accuracy using points. Collaboration is permitted, but all students must turn in their own individual work. These cannot be retaken or revised.
2. **Checkpoints**: Each learning standard will be tested at two levels: Level 1 (basic) or Level 2 (advanced). A solution will either meet the learning standard at the given level or not; there are no points, percentages, or partial credit. A retake will be offered for each checkpoint, without grade penalty, approximately one week after it is returned. The retake will consist of different problems, covering the same learning standards at the same level of difficulty as the original assessment. To be eligible for the retake, students must complete corrections of their original assessment, as well as additional practice.
3. **Portfolio**: The portfolio will be awarded one of three marks: "Does not meet expectations", "Meets expectations", or "Excellent". Instructions for the portfolio will be provided, including specifications indicating the requirements to earn the marks "Meets Expectations" and "Excellent".
4. **Final Exam**: There will be a cumulative final exam covering the 25 learning targets listed above. The questions will be at the same level of difficulty as those on the checkpoints. The final exam grade will be recorded as a percentage.

Assignment of Letter Grades

Final letter grades will be assigned according to the following table:

	C-	C	B	A
Learning standards met at Level 2	5	10	15	20
Learning standards met at Level 1	15	18	20	25
Problem Sets	60%	70%	80%	90%
Portfolio	No requirement	Meets Expectations	Meets Expectations	Excellent

All requirements in a given column must be met in order to receive the corresponding grade. If all requirements for the leftmost column are not met, then a grade of 'D' will be assigned.

+/- Grades:

- If 3/4 of the requirements in the next column are met *or* the final exam score is 85% or above, then a final letter grade of *C* or *B* will be augmented with a '+'.
- If the final exam score is below 50%, then a final letter grade of *C* or *B* will be augmented with a '−', while a final letter grade of *C*+ or *B*+ will be lowered to *C* or *B*, respectively.

Teaching and Learning Philosophy

My approach to teaching the course will focus on developing conceptual understanding, procedural fluency, mathematical reasoning, and problem-solving. All students are strongly encouraged to contribute to class discussions. I encourage everyone to challenge themselves and to step outside of their comfort zone.

Diversity, Equity, and Inclusion Statement

Everyone has the right to be included and treated with the highest level of respect, to be addressed by their preferred name and pronouns, and to have their identity respected. Students are encouraged to provide the instructor with regular input on how they are experiencing the course throughout the semester.

Equity Goals

- Ensure that each student has safe and accessible opportunities and support to be able to enhance their knowledge and understanding of mathematics.
- Provide multiple approaches for learning and assessment to meet the various needs of our diverse group of students and their backgrounds.
- Helping students see the value in developing the ability to have mathematical discussions and to disagree cooperatively in critiquing the reasoning of others, and not adversarially.
- Ensuring that all students attain mathematics proficiency and increasing the numbers of students from all racial, ethnic, linguistic, gender, and socioeconomic groups who attain the highest levels of mathematics achievement. (NCTM 2020)
- Creating, supporting, and sustaining a culture of access and equity that is responsive to students' backgrounds, experiences, cultural perspectives, traditions, and knowledge. (NCTM 2020)

Academic Behavior Code

Students are expected to follow the IMSA 2023–2024 Student Handbook regarding academic honesty and IMSA's Academic Behavior Code.

Learning Standards SSLs

Developing the Tools of Thought
I. IMSA students develop automaticity in skills, concepts, and processes that support and enable complex thought.
II. IMSA students construct questions which further understanding, forge connections, and deepen meaning.
IV. IMSA students evaluate the soundness and relevance of information and reasoning.

Extending and Integrating Thought
II. IMSA students recognize, pursue, and explain substantive connections within and among areas of knowledge.

Expressing and Evaluating Constructs
I. IMSA students construct and support judgments based upon evidence.

II. IMSA students write and speak with power, economy, and elegance.

This syllabus is subject to change throughout the semester as needed at the instructor's discretion.